

Two-Sided Inverse-Gaussian Poisson Square Bounds Characterize UMD

FA Banach run fa_banach_001, agent agent_lane_10

23 July 2026

Status

This packet records a candidate full solution, likely valid, to the question in the remark following Theorem 1.5 of Almeida, Betancor, Fariña, and Rodríguez-Mesa [1]. The answer is affirmative.

Source Passage

$$\leq C \left\| g_{\{P_t\}_{t>0}}^{1,X}(f) \right\|_{L^p(\mathbb{R}^n, dx)} \left\| g_{\{P_t\}_{t>0}}^{1,X^*}(h) \right\|_{L^{p'}_{X^*}(\mathbb{R}^n, dx)} \leq C \left\| g_{\{P_t\}_{t>0}}^{1,X}(f) \right\|_{L^p(\mathbb{R}^n, dx)} \|h\|_{L^{p'}_{X^*}(\mathbb{R}^n, dx)}.$$

Since $L^{p'}_{X^*}(\mathbb{R}^n, dx)$ is norming in $L^p_X(\mathbb{R}^n, dx)$ it follows that

$$\|f\|_{L^p_X(\mathbb{R}^n, dx)} \leq C \left\| g_{\{P_t\}_{t>0}}^{1,X}(f) \right\|_{L^p_X(\mathbb{R}^n, dx)}.$$

From [19] Theorem 1.6] we deduce that X is UMD and the proof is complete.

Remark. The property established in [19] Theorem 1.6] suggests to ask whether in Theorem 1.5 (ii) we can replace the condition involving the dual space X^* by this other one

$$\|f\|_{L^p_X(\mathbb{R}^n, \gamma_{-1})} \leq C \left\| g_{\{P_t^{\mathcal{A}}\}_{t>0}}^{1,X}(f) \right\|_{L^p_X(\mathbb{R}^n, \gamma_{-1})}, \quad f \in L^p_X(\mathbb{R}^n, \gamma_{-1}),$$

and then the UMD property for X can be again deduced from (ii). At this moment we do not know what is the answer for this question.

4.2. **Proof of Theorem 1.6.** In [6] Theorems 1.4 and 1.5] the Banach spaces with the UMD-property were characterized by using $L^p(\mathbb{R}^n, \gamma_{-1})$ -boundedness properties of the Riesz transforms associate with the operator $\mathcal{A}$. For every $i = 1, \dots, n$, the Riesz transform $\mathcal{R}_i^{\mathcal{A}}$ in the $\mathcal{A}$ -setting is defined by

Transcription and Logical Scope of the Question

Let X be an order-continuous Köthe function space, let $1 < p < \infty$, and write

$$\mathcal{G}_{\mathcal{A}}^X f(x, w) = \left(\int_0^\infty |t \partial_t P_t^{\mathcal{A}}(f(\cdot, w))(x)|^2 \frac{dt}{t} \right)^{1/2}.$$

Theorem 1.5(ii) of [1] assumes an upper estimate for $\mathcal{G}_{\mathcal{A}}^X$ on $L^p_X(\mathbb{R}^n, \gamma_{-1})$ and an upper estimate for the dual-space square function on $L^{p'}_{X^*}$. The source asks whether the second, dual-space estimate can instead be replaced by

$$\|f\|_{L^p_X(\mathbb{R}^n, \gamma_{-1})} \leq C \left\| \mathcal{G}_{\mathcal{A}}^X f \right\|_{L^p_X(\mathbb{R}^n, \gamma_{-1})}, \tag{R}$$

and whether UMD then follows.

Thus the modified condition retains the first estimate in Theorem 1.5(ii):

$$\|\mathcal{G}_A^X f\|_{L_X^p(\mathbb{R}^n, \gamma_{-1})} \leq C \|f\|_{L_X^p(\mathbb{R}^n, \gamma_{-1})}. \quad (\text{U})$$

The question is about the two-sided pair (U)–(R), not about (R) alone.

Idea of the Proof

The inverse-Gaussian geometry becomes Euclidean at small scales. Test (U)–(R) on a shrinking bump $f_\varepsilon(x) = F(x/\varepsilon)$. After the changes $x = \varepsilon z$ and $t = \varepsilon s$, the generator becomes

$$B_\varepsilon = -\frac{1}{2}\Delta - \varepsilon^2 z \cdot \nabla,$$

the measure becomes $e^{\varepsilon^2|z|^2} dz$, and B_ε tends to the Euclidean generator $-\Delta/2$.

The required convergence of the full square-function norms is not left to an abstract semigroup limit. On a fixed physical ball, the local comparison estimate proved in [1] gives a rescaled error of size $\varepsilon^{1/2} I_{1/2}(\|F\|_X)$, whose norm tends to zero for every $p > 1$. Outside that ball, the input and output are separated. The explicit global kernel estimate (A2) in [1] supplies an $L^p(\gamma_{-1})$ Gaussian majorant, so the normalized tail is $O(\varepsilon^{n(1-1/p)})$.

Consequently the two inverse-Gaussian inequalities pass to two-sided Euclidean Poisson square-function inequalities. The Xu–Hytönen characterization then gives UMD.

The Tangent Lemma

Let P_t denote the Euclidean Poisson semigroup subordinated to the heat semigroup generated by $-\Delta/2$, and put

$$\mathcal{G}_0^X F(z, w) = \left(\int_0^\infty |t \partial_t P_t(F(\cdot, w))(z)|^2 \frac{dt}{t} \right)^{1/2}.$$

The harmless choice of the factor $1/2$ does not change the square function: a constant rescaling of t leaves dt/t invariant.

Lemma 1 (Small-scale tangent lemma). *Let $1 < p < \infty$, let X be a Köthe function space, and let*

$$F(z, w) = \sum_{j=1}^N \phi_j(z) x_j(w), \quad \phi_j \in C_c^\infty(\mathbb{R}^n), \quad x_j \in X.$$

For $0 < \varepsilon < 1$, define $f_\varepsilon(x, w) = F(x/\varepsilon, w)$. If $d\gamma_{-1}(x) = c_n e^{|x|^2} dx$, then

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{-n/p} \|f_\varepsilon\|_{L_X^p(\gamma_{-1})} = c_n^{1/p} \|F\|_{L_X^p(dz)} \quad (1)$$

and

$$\lim_{\varepsilon \downarrow 0} \varepsilon^{-n/p} \|\mathcal{G}_A^X f_\varepsilon\|_{L_X^p(\gamma_{-1})} = c_n^{1/p} \|\mathcal{G}_0^X F\|_{L_X^p(dz)}. \quad (2)$$

Proof. The first limit follows immediately from

$$\varepsilon^{-n} \|f_\varepsilon\|_{L_X^p(\gamma_{-1})}^p = c_n \int_{\mathbb{R}^n} \|F(z)\|_X^p e^{\varepsilon^2|z|^2} dz$$

and dominated convergence, since F is compactly supported.

We prove (2). Choose $R > 0$ with $\text{supp } F \subset B(0, R)$, and then choose

$$0 < \rho < \min\{1, n/4\}.$$

For all sufficiently small ε , if $|x| < \rho$ and $y \in \varepsilon B(0, R)$, then (x, y) belongs to the local region N_1 of [1]. The local comparison estimate proved there is

$$\|t\partial_t(P_t^{\mathcal{A}}(x, y) - P_t(x - y))\|_{L^2((0, \infty), dt/t)} \leq C \frac{\sqrt{1 + |x|}}{|x - y|^{n-1/2}}. \quad (3)$$

The reverse triangle inequality in $L^2(dt/t)$, followed by Minkowski's inequality in the lattice, therefore gives, for $|z| < \rho/\varepsilon$,

$$\begin{aligned} & \|\mathcal{G}_{\mathcal{A}}^X f_{\varepsilon}(\varepsilon z) - \mathcal{G}_0^X F(z)\|_X \\ & \leq C\varepsilon^{1/2} \int_{B(0, R)} \frac{\|F(u)\|_X}{|z - u|^{n-1/2}} du =: C\varepsilon^{1/2} J(z). \end{aligned} \quad (4)$$

Here we also used the exact Euclidean scaling $\mathcal{G}_0^X f_{\varepsilon}(\varepsilon z) = \mathcal{G}_0^X F(z)$.

The function J is bounded on bounded sets because $|v|^{-n+1/2}$ is locally integrable, while

$$J(z) \leq C_F |z|^{-n+1/2} \quad (|z| > 2R).$$

Moreover, $e^{\varepsilon^2|z|^2} \leq e^{\rho^2}$ on the ball under consideration. Hence the p -th power of the norm in (4), with the rescaled weight, is bounded by a constant times

$$\varepsilon^{p/2} \left(1 + \int_1^{\rho/\varepsilon} r^{n-1-p(n-1/2)} dr \right). \quad (5)$$

If $p(n - 1/2) > n$, this is $O(\varepsilon^{p/2})$. At equality it is $O(\varepsilon^{p/2} \log(1/\varepsilon))$. If $p(n - 1/2) < n$, it is $O(\varepsilon^{n(p-1)})$. Thus (4) tends to zero in $L_X^p(B(0, \rho/\varepsilon), e^{\varepsilon^2|z|^2} dz)$ for every $p > 1$.

It remains to show that the inverse-Gaussian output outside the fixed physical ball is negligible after normalization. Define the scalar square kernel

$$K_{\mathcal{A}}(x, y) = \|t\partial_t P_t^{\mathcal{A}}(x, y)\|_{L^2((0, \infty), dt/t)}.$$

We claim that, after decreasing the allowed size of y if necessary, there is a function $q \in L^p(\mathbb{R}^n, \gamma_{-1})$ such that

$$K_{\mathcal{A}}(x, y) \leq q(x), \quad |x| \geq \rho, \quad |y| \leq \rho/2. \quad (6)$$

On a fixed annulus $\rho \leq |x| \leq R_0$, this follows from (3) together with the Euclidean bound

$$\|t\partial_t P_t(x - y)\|_{L^2(dt/t)} \leq C|x - y|^{-n}$$

on the local part, and from the source global bound on the complementary part. All pairs are uniformly separated there.

For R_0 sufficiently large, every pair with $|x| > R_0$ and $|y| \leq \rho/2$ is global. The Poisson-to-heat comparison in [1], followed by its estimate (A2) with $m = 1, k = 0, q = 2$, gives parameters $\theta, \delta > 0$, chosen so that $\theta + \delta > 2/p$, and the two bounds

$$K_{\mathcal{A}}(x, y) \leq C e^{-(\theta+\delta)|x|^2/2 + (\theta-\delta)|y|^2/2} \quad \text{if } \langle x, y \rangle \leq 0, \quad (7)$$

and

$$K_{\mathcal{A}}(x, y) \leq C|x + y|^n e^{\theta(|y|^2 - |x|^2)/2 - \delta|x+y||x-y|/2} \quad \text{if } \langle x, y \rangle > 0. \quad (8)$$

Since

$$|x + y||x - y| \geq ||x|^2 - |y|^2|,$$

both cases, uniformly for bounded y , are dominated for large x by

$$C(1 + |x|)^n e^{-a|x|^2} \quad \text{for some } a > 1/p.$$

This belongs to $L^p(\gamma_{-1})$, proving (6).

Let $M_F = \int_{B(0,R)} \|F(u)\|_X \, du$. Minkowski and (6) give, for small ε and $|x| \geq \rho$,

$$\|\mathcal{G}_{\mathcal{A}}^X f_{\varepsilon}(x)\|_X \leq \int_{\varepsilon B(0,R)} K_{\mathcal{A}}(x, y) \|F(y/\varepsilon)\|_X \, dy \leq \varepsilon^n M_F q(x).$$

Consequently,

$$\varepsilon^{-n/p} \|\mathcal{G}_{\mathcal{A}}^X f_{\varepsilon}\|_{L_X^p(\{|x| \geq \rho\}, \gamma_{-1})} \leq M_F \|q\|_{L^p(\gamma_{-1})} \varepsilon^{n(1-1/p)} \longrightarrow 0. \quad (9)$$

Finally, $\mathcal{G}_0^X F \in L_X^p(\mathbb{R}^n)$: it is dominated by a finite sum of scalar Poisson square functions times $\|x_j\|_X$. On $|z| < \rho/\varepsilon$, the weight is bounded by e^{ρ^2} , converges pointwise to one, and the domains increase to $\mathbb{R}^n$. Dominated convergence therefore yields

$$\int_{|z| < \rho/\varepsilon} \|\mathcal{G}_0^X F(z)\|_X^p e^{\varepsilon^2|z|^2} \, dz \longrightarrow \|\mathcal{G}_0^X F\|_{L_X^p(\mathbb{R}^n)}^p.$$

Combine this with the local error estimate and (9), then undo the change $x = \varepsilon z$, to obtain (2). $\square$

Full Answer

Theorem 1. *Let X be an order-continuous Köthe function space and $1 < p < \infty$. The following are equivalent.*

1. X is UMD.
2. There is a constant C such that, for every $f \in L_X^p(\mathbb{R}^n, \gamma_{-1})$,

$$C^{-1} \|f\|_{L_X^p(\gamma_{-1})} \leq \|\mathcal{G}_{\mathcal{A}}^X f\|_{L_X^p(\gamma_{-1})} \leq C \|f\|_{L_X^p(\gamma_{-1})}.$$

In particular, the proposed replacement of the X^ -condition in Theorem 1.5(ii) of [1] is valid.*

Proof. Assume the two inverse-Gaussian estimates. Apply them to the functions f_{ε} in Lemma 1, divide by $\varepsilon^{n/p}$, and let $\varepsilon \downarrow 0$. Equations (1) and (2) give, on compactly supported finite tensors,

$$C^{-1} \|F\|_{L_X^p(\mathbb{R}^n)} \leq \|\mathcal{G}_0^X F\|_{L_X^p(\mathbb{R}^n)} \leq C \|F\|_{L_X^p(\mathbb{R}^n)}. \quad (10)$$

The upper estimate and the reverse triangle inequality for the $L^2(dt/t)$ norm extend the square-function map continuously from this dense test class to all of $L_X^p(\mathbb{R}^n)$; the lower estimate passes to the completion as well.

Xu's Banach-lattice theorem [3, Theorem 4.1], equivalently the Banach-lattice specialization of Hytönen's characterization [2, Theorem 1.6], says that the two-sided Euclidean Poisson lattice-square estimate (10) implies that X is UMD. This proves the direction asked in the source remark.

Conversely, if X is UMD, then the two-sided inverse-Gaussian estimate is the UMD implication for subordinated symmetric diffusion semigroups recalled as Theorem D(i) in [1]. Hence the conditions are equivalent. $\square$

Verification Notes

The proof uses only the local comparison estimate, the Poisson-to-heat comparison, and the explicit global estimate (A2) already proved in the source paper, followed by the classical Euclidean two-sided-square characterization of UMD.

The local radial integral (5) was checked separately in all three exponent regimes. The far-tail exponent is strict because the parameters in (A2) are selected in the source with $\theta + \delta > 2/p$. The order-continuity assumption is exactly the source assumption; no stronger lattice hypothesis is introduced.

No numerical experiment is used. A separate line-by-line audit is included as `verification.md`.

Novelty and Literature Check

The lightweight run indexes were searched for arXiv:2102.13381, the paper title, inverse-Gaussian square functions, reverse estimates, UMD, and removal of the dual-space hypothesis. No existing result packet answers this exact question.

Bounded external searches on 23 July 2026 used the exact phrase “replace the condition involving the dual space”, the full title together with “UMD reverse estimate”, and close inverse-Gaussian square-function formulations. They returned the source paper and its Potential Analysis publication, but no later paper explicitly resolving the remark. The open-access published version was also inspected directly: the question remains verbatim on published PDF page 43, so it was not removed or answered during peer review. This supports, but cannot guarantee, novelty.

Limitations and Human-Review Recommendation

The theorem answers the source question exactly. It does not assert that the reverse estimate (R) alone implies UMD; the retained upper estimate (U) is essential to the statement being solved.

Human review should focus on Lemma 1, especially the uniform majorant (6) extracted from (A2). The packet spells out both sign cases of (A2), the required strict exponent, the compact off-diagonal region, and the normalization of the shrinking tail.

References

- [1] V. Almeida, J. J. Betancor, J. C. Fariña, and L. Rodríguez-Mesa, *Littlewood-Paley-Stein theory and Banach spaces in the inverse Gaussian setting*, Potential Analysis 59 (2023), 1235–1284; arXiv:2102.13381.
- [2] T. P. Hytönen, *Littlewood-Paley-Stein theory for semigroups in UMD spaces*, Rev. Mat. Iberoam. 23 (2007), no. 3, 973–1009.
- [3] Q. Xu, *Littlewood-Paley theory for functions with values in uniformly convex spaces*, J. Reine Angew. Math. 504 (1998), 195–226.